CHENNAI INSTITUTE OF TECHNOLOGY
(Autonomous)

(An Autonomous Institution, Affiliated to Anna University, Chennai)

NAME OF THE PROGRAMME (Mention, if common)	**BE., COMPUTER SCIENCE AND ENGINEERING**		**LEVEL : UG**
YEAR / SEMESTER	**II/IV**		
REGULATIONS	**2024**		
COURSE CODE	**CS4403**		
COURSE NAME	**MACHINE LEARNING**		
FACULTY NAME (Prepared by)	**Ms. POORNIMA LAKSHMI**	**Contact Number**	**8870964537**
NAME OF THE SUBJECT EXPERT(Verified by)	**Dr. P. AURCHANA**	**Contact Number**	**7603855239**

REVISED BLOOMS TAXONOMY(RBT) :

L1- Remembering	**L2 - Understanding**	**L3 - Applying**	**L4 - Analyzing**	**L5 - Evaluating**	**L6 - Creating**

UNIT-I: INTRODUCTION

Introduction to Machine Learning: Introduction. Different types of learning, Hypothesis space and inductive bias, Evaluation. Training and test sets, cross validation, Concept of over fitting, under fitting, Bias and Variance. Linear Regression: Introduction, Linear regression, Simple and Multiple Linear regression, Polynomial regression, evaluating regression fit.

PART-A (2-Marks)

S.NO	QUESTIONS	CO	RBT LEVEL
1	Illustrate Machine Learning.	CO1	L1
2	List any two real-world applications of Machine Learning.	CO1	L1
3	Explain supervised learning? Give one example	CO1	L1
4	Elaborate unsupervised learning.	CO1	L1
5	Discuss about reinforcement learning.	CO1	L2
6	Explain the term hypothesis in Machine Learning?	CO1	L1

7	Elaborate hypothesis space.	**CO1**	**L1**
8	Explain the term inductive bias with an example.	**CO1**	**L2**
9	Discuss the training data with a neat example.	**CO1**	**L2**
10	Define test dataset.	**CO1**	**L1**
11	Write in detail about cross-validation.	**CO1**	**L1**
12	Elaborate overfitting.	**CO1**	**L1**
13	Illustrate underfitting.	**CO1**	**L1**
14	Define bias in Machine Learning models.	**CO1**	**L1**
15	Define variance in Machine Learning.	**CO1**	**L1**
16	State the bias–variance trade-off.	**CO1**	**L2**
17	Define linear regression.	**CO1**	**L1**
18	Give the correct explanation for simple linear regression?	**CO1**	**L1**
19	Explain about multiple linear regression?	**CO1**	**L1**
20	Define polynomial regression and also define best fit line.	**CO1**	**L1**

PART-B (13 - Marks)			**RBT Level**	**Marks**
1	Elaborate Machine Learning and its importance in modern computing.	**CO1**	**L2**	**13**
2	Describe different types of Machine Learning with suitable examples.	**CO1**	**L2**	**13**
3	Analyze the impact of model complexity on regression performance.	**CO1**	**L4**	**13**
4	Evaluate the performance of a regression model using suitable metrics.	**CO1**	**L4**	**13**
5	Design a regression model for a real-world problem and justify the choice.	**CO1**	**L4**	**13**
6	Analyze a case where linear regression fails and explain why.	**CO1**	**L4**	**13**
7	Illustrate a real-world application of Machine Learning using regression	**CO1**	**L3**	**13**
8	Propose a Machine Learning approach to solve a prediction problem.	**CO1**	**L5**	**13**
9	Critically analyze the limitations of linear regression models.	**CO1**	**L4**	**13**
10	Develop a Machine Learning workflow from data collection to evaluation.	**CO1**	**L5**	**13**

11	Compare supervised, unsupervised, and reinforcement learning.	**CO1**	**L3**	**13**
12	Explain the concept of hypothesis space and inductive bias in detail.	**CO1**	**L2**	**13**
13	Analyze the role of inductive bias in improving model generalization.	**CO1**	**L4**	**13**
14	Explain training set, test set, and validation set with a neat diagram.	**CO1**	**L2**	**13**
15	Discuss the need for model evaluation in Machine Learning.	**CO1**	**L2**	**13**
16	Explain cross-validation techniques in detail.	**CO1**	**L2**	**13**
17	Analyze how cross-validation helps prevent overfitting.	**CO1**	**L2**	**13**
18	Design a suitable evaluation strategy for a Machine Learning model.	**CO1**	**L5**	**13**
19	Analyze the bias–variance trade-off in Machine Learning.	**CO1**	**L2**	**13**
20	Compare overfitting and underfitting in Machine Learning models.	**CO1**	**L2**	**13**
PART-C (15 - Marks)		**CO**	**RB2 T Level**	**Marks**
1	Describe the complete workflow of a machine learning system. Include data splitting, model training, evaluation, and validation	**CO1**	**L2**	**15**
2	Explain polynomial regression and its significance.	**CO1**	**L2**	**15**
3	Discuss methods used to evaluate regression models.	**CO1**	**L2**	**15**
4	Explain Mean Squared Error (MSE) and R^2 score.	**CO1**	**L3**	**15**
5	Explain multiple linear regression with an example.	**CO1**	**L3**	**15**

UNIT-2: SUPERVISED LEARNING			
Bayesian linear regression, gradient descent, Linear Classification Models: Discriminant function – Perceptron algorithm, Probabilistic discriminative model - Logistic regression, Probabilistic generative model - Naïve Bayes, Maximum margin classifier – Support vector machine, Decision Tree, Random Forests, Instance Based Learning- KNN.			
PART-A (2-Marks)			
S.NO	**QUESTIONS**	**CO**	**RBT LEVEL**
1	Write the types of Gradient descent and differentiate it.	**CO2**	**L1**

2	How Regression differs from classification?	**CO2**	**L1**
3	Write the sigmoid function used in logistic regression.	**CO2**	**L1**
4	A linear SVM has weight vector w=[1,−1] and bias b =2. Compute the decision function for the point x = [3,1] and predict the class	**CO2**	**L1**
5	A Random Forest has 5 decision trees. For a data point, the trees predict : [+1, -1, +1, +1, -1]. What is the final predicted class using majority voting?	**CO2**	**L1**
6	For a KNN regression problem, given the 3 nearest neighbors' target values are [12, 15, 20], compute the predicted value using the average method.	**CO2**	**L1**
7	List out the advantages of SVM	**CO2**	**L1**
8	How does random forest reduce overfitting compared to a single decision tree?	**CO2**	**L1**
9	Develop the concepts of K- Nearest Neighbours.	**CO2**	**L2**
10	How does the structure of decision tree help in classifying a data instance?	**CO2**	**L2**
11	State Bayes theorem**.**	**CO2**	**L2**
12	Define maximum margin classifier**.**	**CO2**	**L2**
13	Discuss the output equation of perceptron algorithm.	**CO2**	**L1**
14	Give the difference between a decision tree and a random forest?	**CO2**	**L1**
15	Identify pros and cons of regression models.	**CO2**	**L1**
16	Discuss about prior distribution in Bayesian learning?	**CO2**	**L2**
17	A logistic regression model gives probability P(y=1|x) = 0.7. How would you choose a threshold to balance precision and recall?	**CO2**	**L1**
18	List the advantage of instance based approach	**CO2**	**L2**
19	Specify the role of kernels? List out the different type of Kernel.	**CO2**	**L1**
20	If the dataset has 100 features, how many features are typically considered at each split in classification?	**CO2**	**L2**

PART-B (13- Marks)		**CO**	**RBT Level**	**Marks**
1	Outline the concept of Random forest? Explain with example?	**CO2**	**L2**	**13**

<table>
<tr><td>2</td><td>Compare logistic regression with Naïve Bayes for a real-time classification problem.</td><td>CO2</td><td>L2</td><td>13</td></tr>
<tr><td>3</td><td>List example of Instance-based learning algorithm.</td><td>CO2</td><td>L2</td><td>13</td></tr>
<tr><td>4</td><td>Consider a Random Forest classifier with 3 decision trees. The trees predict the following classes for a data point:
Tree 1 → Class A
Tree 2 → Class B
Tree 3 → Class A
a) What is the final prediction using majority voting?
b) Explain the advantages of using a Random Forest over a single decision tree.
c) What role does randomness in features and bagging play in Random Forests?</td><td>CO2</td><td>L2</td><td>13</td></tr>
<tr><td>5</td><td>Differentiate linear regression and logistic regression</td><td>CO2</td><td>L2</td><td>13</td></tr>
<tr><td>6</td><td>Discuss in detail about Perceptron algorithm.</td><td>CO2</td><td>L2</td><td>13</td></tr>
<tr><td>7</td><td>Describe how Bayesian linear regression handles small datasets compared to classical linear regression.</td><td>CO2</td><td>L2</td><td>13</td></tr>
<tr><td>8</td><td>Describe the difference between linear and non-linear SVM.</td><td>CO2</td><td>L2</td><td>13</td></tr>
<tr><td>9</td><td>Describe the role of entropy and information gain in constructing decision trees. How do these measures influence the splitting of nodes during the tree-building process?</td><td>CO2</td><td>L3</td><td>13</td></tr>
<tr><td>10</td><td>Given the following dataset of 2D points with labels:
<table>
<tr><th>X</th><th>Y</th><th>Label</th></tr>
<tr><td>1</td><td>2</td><td>A</td></tr>
<tr><td>2</td><td>3</td><td>A</td></tr>
<tr><td>3</td><td>3</td><td>B</td></tr>
<tr><td>5</td><td>4</td><td>B</td></tr>
</table>
Predict the label of the point (3,2) using KNN with k=3 using Euclidean distance</td><td>CO2</td><td>L3</td><td>13</td></tr>
<tr><td>11</td><td>A linear SVM has weights w = [2,3] and bias b = -5. Compute the decision function for the point x = [1,2] and predict the class.</td><td>CO2</td><td>L3</td><td>13</td></tr>
<tr><td>12</td><td>Discuss how Random Forest handles missing values and high-dimensional datasets.</td><td>CO2</td><td>L3</td><td>13</td></tr>
<tr><td>13</td><td>Using KNN, determine which sport does Jaffrin belongs to:
<table>
<tr><th>Name</th><th>Age</th><th>Gender</th><th>Sport</th></tr>
<tr><td>Dev</td><td>32</td><td>M</td><td>football</td></tr>
<tr><td>Mark</td><td>40</td><td>M</td><td>Neither</td></tr>
<tr><td>Zara</td><td>16</td><td>F</td><td>Cricket</td></tr>
</table></td><td>CO2</td><td>L3</td><td>13</td></tr>
</table>

		Zaira	34	F	Cricket			
		Sachin	55	M	Neither			
		Rahul	40	M	Cricket			
		Pooja	20	F	Neither			
		Smith	15	M	Cricket			
		Lakshmi	55	F	Football			
		Michael	15	M	Football			
14	Derive formulas for entropy and Gini Index. Given a small dataset, compute entropy, information gain, and Gini Index for each feature. Construct a complete decision tree manually.					**CO2**	**L3**	**13**
15	Calculate the Euclidean distance between the points (2,3,5) and (5,1,7) in 3D space.					**CO2**	**L3**	**13**
16	A dataset of email messages (spam=1, not spam=0). Using logistic regression, calculate the log-likelihood for the following prediction: a)True label y = 1, predicted probability p = 0.8 b) True label y = 0, predicted probability p = 0.3					**CO2**	**L3**	**13**
17	Show how posterior probability is computed using Naïve Bayes classifier.					**CO2**	**L3**	**13**
18	For a KNN regression problem, given the 3 nearest neighbors' target values are [12, 15, 20], compute the predicted value using the average method.					**CO2**	**L3**	**13**
19	A bank wants to predict whether a customer will be approved for a loan YES = 0 , NO = 1. X1 = Credit Score ;X2 = Monthly Income ; W1 = -0.005; W2 = -0.04; Bias = 4. A new customer has a credit score of 650 and income of Rs. 50000, the threshold value is 0.5. Solve by using Logistic Regression.					**CO2**	**L3**	**13**
20	Classify "OR GATE" using Perceptron algorithm: w1=1.2, w2=0.6;Threshold = 1; Learning Rate = 0.5					**CO2**	**L3**	**13**
PART-C (15-Marks)						**CO**	**RBT Level**	**Marks**

<table>
<tr><td>1</td><td>Consider the training dataset given in the following table. Use Weighted k-NN and determine the class. Test instance (7.6, 60, 8) and K=3.
<table>
<tr><th>S.No.</th><th>CGPA</th><th>Assessment</th><th>Project Submitted</th><th>Result</th></tr>
<tr><td>1</td><td>9.2</td><td>85</td><td>8</td><td>Pass</td></tr>
<tr><td>2</td><td>8</td><td>80</td><td>7</td><td>Pass</td></tr>
<tr><td>3</td><td>8.5</td><td>81</td><td>8</td><td>Pass</td></tr>
<tr><td>4</td><td>6</td><td>45</td><td>5</td><td>Fail</td></tr>
<tr><td>5</td><td>6.5</td><td>50</td><td>4</td><td>Fail</td></tr>
<tr><td>6</td><td>8.2</td><td>72</td><td>7</td><td>Pass</td></tr>
<tr><td>7</td><td>5.8</td><td>38</td><td>5</td><td>Fail</td></tr>
<tr><td>8</td><td>8.9</td><td>91</td><td>9</td><td>Pass</td></tr>
</table></td><td>CO2</td><td>L3</td><td>15</td></tr>
<tr><td>2</td><td>Illustrate Support Vector Machine (SVM) with: Maximum margin concept, Hyperplane, Support vectors.</td><td>CO2</td><td>L3</td><td>15</td></tr>
<tr><td>3</td><td>Classify “AND GATE” using Perceptron algorithm: w1=1.2, w2=0.6;Threshold = 1; Learning Rate = 0.5</td><td>CO2</td><td>L3</td><td>15</td></tr>
<tr><td>4</td><td>Classify whether a person will Play Tennis (Yes/No) based on the weather using Naïve’s Bayes Theorem
<table>
<tr><td>Sunny</td><td>No</td></tr>
<tr><td>Sunny</td><td>No</td></tr>
<tr><td>Overcast</td><td>Yes</td></tr>
<tr><td>Rain</td><td>Yes</td></tr>
<tr><td>Rain</td><td>Yes</td></tr>
<tr><td>Rain</td><td>No</td></tr>
<tr><td>Overcast</td><td>Yes</td></tr>
<tr><td>Sunny</td><td>No</td></tr>
<tr><td>Sunny</td><td>Yes</td></tr>
<tr><td>Rain</td><td>Yes</td></tr>
</table></td><td>CO2</td><td>L3</td><td>15</td></tr>
<tr><td>5</td><td>Compare decision tree and random forest in terms of bias, variance, and performance.</td><td>CO2</td><td>L</td><td>15</td></tr>
</table>

UNIT-3: NEURAL NETWORKS			
Artificial Neural Networks: Introduction, Biological motivation, ANN representation, appropriate problem for ANN learning, Perceptron, multilayer networks and the back propagation algorithm, Popular CNN Architectures, RNNs, LSTM, BERT, GANS and Generative Models.			
PART-A (2-Marks)			
S.NO	**QUESTIONS**	**CO**	**RBT LEVEL**
1	State the Perceptron rule.	**CO3**	**L1**
2	Formulate the meaning behind Artificial Neural Network	**CO3**	**L1**
3	How to evaluate the update weight in back propagation?	**CO3**	**L1**
4	Clarify Stride	**CO3**	**L1**
5	Give the meaning of activation function and its importance in neural network?	**CO3**	**L2**
6	Mention the pros and cons of bidirectional RNN.	**CO3**	**L1**
7	Specify the types of GAN?	**CO3**	**L1**
8	Clarify the concept of Recurrent Neural Network	**CO3**	**L2**
9	Give the meaning of BERT?	**CO3**	**L1**
10	State Generative Model.	**CO3**	**L2**
11	Compare generator and discriminator.	**CO3**	**L2**
12	Compute Optimal solution for the given data using GAN : Pdata (x) = 0.5 ; Pg(x) = 0.3	**CO3**	**L2**
13	Compute Discriminator Loss: D(x)= 0.7;D(G(Z)) = 0.2	**CO3**	**L2**
14	Give the meaning of Convolutional Neural Network	**CO3**	**L1**

15	Give the main limitation of RNN?	CO3	L2
16	Why BERT is Important	CO3	L2
17	Give the meaning of Leaky units?	CO3	L2
18	Write short notes on vanishing Gradient problem.	CO3	L2
19	Distinguish neural network and recurrent neural network	CO3	L2
20	State the biological motivation behind ANN.	CO3	L2

	PART-B (13- Marks)	CO	RBT Level	Marks
1	Discuss about the fully connected layers in detail.	CO3	L2	13
2	Formulate the LSTM architecture and specify its main components.	CO3	L2	13
3	Solve by using LSTM at time step t Current input: $X_t = 1$; $h_{t-1}=0.5$; $C_{t-1}=0.2$. Find O_t and h_t using activation function sigmoid and tanh. GATE / WEIGHT / BIAS Forget gate / 0.7 / 0.1 Input gate / 0.6 / 0.2 Candidate Cell / 0.9 / 0 Output Gate / 0.5 / 0.1	CO3	L3	13
4	Give the suitable GANs with applications.	CO3	L2	13
5	Compare CNN and RNN	CO3	L2	13
6	Provide the concept of a Perceptron with a neat diagram	CO3	L3	13
7	Derive the Backpropagation rule considering the training rule for Output Unit weights and Training Rule for Hidden Unit weights with example.	CO3	L3	13
8	Discuss RNN with an example.	CO3	L3	13
9	Elaborate Multilayer Neural Networks and the need for Hidden Layers.	CO3	L2	13
10	Elaborate the architecture and learning process of Perceptron. Discuss its limitations	CO3	L2	13
11	Interpret the Back Propagation algorithm with necessary equations.	CO3	L3	13

12	Discuss the working of the Backpropagation algorithm in detail.Describe forward propagation, error calculation, backward propagation, and weight updates with mathematical expressions.	**CO3**	**L2**	**13**
13	Illustrate simple model of an Artificial Neuron and its functions.	**CO3**	**L2**	**13**
14	Consider simple RNN with one input neuron, one hidden neuron and neutron Given : input sequence : x1 = 1 ; x2 = 2 Initial hidden state = 1 ; w(xh) = 1 ; w(hh) = 1; w(hy)=1; Bias : bh=0 ; by = 0 ; activation Function = tanh	**CO3**	**L3**	**13**
15	Explain the sequence to sequence RNN in detail.	**CO3**	**L3**	**13**
16	Discuss Generative Adversarial Networks (GANs) with generator and discriminator.	**CO3**	**L3**	**13**
17	Specify the concept of generative Adversarial Networks in detail.	**CO3**	**L2**	**13**
18	Discuss the BERT architecture in detail with neat diagram	**CO3**	**L2**	**13**
19	An RGB image of size 64 × 64 × 3 is passed through a CNN with: Convolution layer: 3×3 filter, stride = 1, padding = 1 Max pooling layer: 2×2, stride = 2 (a) Calculate the output feature map size after convolution. (b) Calculate the output size after pooling. (c) Explain how CNN reduces parameters compared to fully connected networks. (d) Calculate the CNN Dimensionality Calculation for the above Numerical Problem	**CO3**	**L3**	**13**
20	Elaborate the gated architecture of LSTM with equations	**CO3**	**L3**	
	PART-C (15-Marks)	**CO**	**RBT Level**	**Marks**
1	Give the detailed description about CNN architecture with diagram.	**CO3**	**L3**	**15**
2	Design single layer perceptron with two iteration. Consider the perceptron having with the initial weights w1=0.5, w2 = 0, learning rate α=0.2 and bias$^{\theta}$ =0.4 for **OR** Boolean function. The activation function is the Step function f(x) which gives the output either 0 or 1. If value of f(x) is greater than or equal to 0, it outputs 1 or else it outputs 0.	**CO3**	**L3**	**15**
3	Analyze in detail the architecture of a Multi- layer Perceptron (MLP). Discuss how the forward and backward passes work, including the role of weights, biases, and activation functions. Provide an example to illustrate the entire process.	**CO3**	**L4**	**15**

4	Provide a detailed explanation of supervised learning. Discuss the components involved in supervised learning models, the process of training these models, and how they are evaluated. Include examples of both classification and regression tasks.	**CO3**	**L4**	**15**
5	An RGB image of size 5 × 5 is passed through a CNN with: Convolution layer: 3×3 filter, stride = 1, padding = 0 Max pooling layer: 2×2, stride = 1. Solve by using CNN. $\text{Input} = \begin{matrix} 1 & 0 & 2 & 3 & 1 \\ 4 & 6 & 6 & 8 & 4 \\ 3 & 1 & 1 & 0 & 2 \\ 1 & 2 & 2 & 4 & 1 \\ 2 & 0 & 1 & 3 & 2 \end{matrix}$	**CO3**	**L3**	**15**

UNIT- 4: UNSUPERVISED LEARNING AND OPTIMISATION			
Unsupervised learning: Expectation maximization - Gaussian mixture models -K-means / K medoid - hierarchical clustering-top-down, bottom-up –single linkage-multiple linkage. Dimensionality Reduction- Linear Discriminant Analysis, Principal Components Analysis, Factor Analysis, Independent Component Analysis. Optimization- Going Downhill, Least-Squares optimization, Conjugate Gradients - Exploitation and Exploration.			
PART-A (2-Marks)			
S.NO	**QUESTIONS**	**CO**	**RBT LEVEL**
1	State the principal of EM Algorithm	**CO4**	**L2**
2	What is the difference between K-means and KNN?	**CO4**	**L1**
3	Sketch the Principal Component Analysis	**CO4**	**L1**
4	Outline the significance Nearest Neighbour methods.	**CO4**	**L1**
5	Distinguish between LDA and PCA.	**CO4**	**L2**
6	Outline about Factor Analysis.	**CO4**	**L1**
7	Classify the different ways for Dimensionality Reduction.	**CO4**	**L2**
8	What is the difference between centroid and medoid?	**CO4**	**L1**
9	Define Conjugate Gradient method.	**CO4**	**L2**

10	What is least squares optimization?	**CO4**	**L1**
11	Define Independent Component Analysis (ICA).	**CO4**	**L1**
12	Define Gaussian Mixture Model	**CO4**	**L1**
13	Define the concepts of Exploitation and Exploration	**CO4**	**L1**
14	Define single linkage in hierarchical clustering.	**CO4**	**L1**
15	What is soft clustering?	**CO4**	**L1**
16	Define least squares optimization.	**CO4**	**L2**
17	What is Independent Component Analysis used for?	**CO4**	**L1**
18	Define cluster distance matrix.	**CO4**	**L2**
19	What is a dendrogram?	**CO4**	**L1**
20	Given covariance matrix: $\begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix}$ Find eigenvalues and identify principal component.	**CO4**	**L2**

	PART-B (13- Marks)	**CO**	**RBT Level**	**Marks**
1	Calculate the Gaussian Mixture Model for the given 1D data where x {2,3,8). Two Gaussian components: Component 1: $\mu_1 = 2$, $\sigma_1^2 = 1$, $\pi_1 = 0.5$ Component 2: $\mu_2 = 8$, $\sigma_2^2 = 1$, $\pi_2 = 0.5$ a) Compute likelihood of each point under both Gaussians. b) Compute responsibilities (γ values).	**CO4**	**L3**	**13**
2	The Dataset contains: (1,2), (2,1), (4,5), (5,4), (10,10). K = 2. Initial centroids: C1 = (1,2), C2 = (10,10). i) Perform two full iterations of K-means. ii) Compute SSE after each iteration. iii) Does algorithm converge? iv) Explain why the point (10,10) influences centroid movement.	**CO4**	**L3**	**13**
3	Two-armed bandit: Arm A mean reward = 5 Arm B mean reward = 4 (i) Define regret.	**CO4**	**L3**	13

	(ii) If always, choose A after 1 trial, what risk exists? (iii) Why ε-greedy reduces regret **(iv)** Use Exploration and Exploitation, and check whether it balances or not			
4	Compare bottom-up and top-down clustering with algorithmic steps and complexity analysis.	**CO4**	**L2**	13
5	The Dataset contains : Dataset: (1,1), (2,2), (3,3), (100,100) where K=2. i) Perform one iteration. ii) Compute centroids. iii) Comment on centroid sensitivity. iv) Would K-medoid change behavior? Why?	**CO4**	**L3**	13
6	Calculate the Principal Component Analysis — 3D to 2D Reduction Dataset: (2,0,0), (0,2,0), (0,0,2) Compute mean and covariance matrix. Identify eigenvalues and Determine number of components to retain 95% variance	**CO4**	**L3**	**13**
7	Perform divisive clustering using Euclidean distance and explain split criterion.	**CO4**	**L2**	**13**
8	Compare and contrast K-Means and K-Medoid clustering algorithms. Discuss their objective functions, computational complexity, advantages, and limitations with suitable examples	**CO4**	**L2**	**13**
9	Calculate the Least Squares Optimization and Gradient Descent for the given the dataset. X: 1, 2, 3 Y: 2, 4, 5 a) Write the Least Square Cost Function b) Computer the Gradient of the cost function with respect to W and b. c) Perform one-step of gradient with learning rate = 0.01, assuming initial w=0, b=0. d) Comment on the role of exploration and exploitation in choosing learning rate	**CO4**	**L3**	**13**
10	Explain difference between hard and soft clustering.	**CO4**	**L3**	**13**
11	Explain how to decide optimal number of clusters using dendrogram.	**CO4**	**L3**	**13**
12	Compare K-means and GMM with mathematical formulation.	**CO4**	**L2**	**13**
13	Derive K-means objective function and show how centroid update minimizes SSE.	**CO4**	**L3**	**13**
14	Calculate LDA for the given two classes: Class 1: (2,3), (3,3) Class 2: (6,5), (7,6)	**CO4**	**L3**	**13**

<table>
<tr><td></td><td>a) Compute class means
b) Compute within-class scatter matrix
c) Compute between-class scatter matrix</td><td></td><td></td><td></td></tr>
<tr><td>15</td><td>Explain Single, Complete and Average linkage methods with examples.</td><td>CO4</td><td>L3</td><td>13</td></tr>
<tr><td>16</td><td>Explain Agglomerative and Divisive hierarchical clustering with linkage criteria.</td><td>CO4</td><td>L3</td><td>13</td></tr>
<tr><td>17</td><td>Explain Gaussian Mixture Models and derive the likelihood function.</td><td>CO4</td><td>L3</td><td>13</td></tr>
<tr><td>18</td><td>Explain Linear Discriminant Analysis with scatter matrices</td><td>CO4</td><td>L3</td><td>13</td></tr>
<tr><td>19</td><td>Explain Conjugate Gradient Method and its advantages.</td><td>CO4</td><td>L3</td><td>13</td></tr>
<tr><td>20</td><td>Explain Least Squares Optimization and derive the normal equation.</td><td>CO4</td><td>L3</td><td>13</td></tr>
<tr><td colspan="2">PART-C (15-Marks)</td><td>CO</td><td>RBT Level</td><td>Marks</td></tr>
<tr><td>1</td><td>Justify the need of Dimensionality Reduction by explaining the following methods
(i) Linear Discriminant Analysis
(ii) Principal Component Analysis
(iii) Independent Component Analysis</td><td>CO4</td><td>L4</td><td>15</td></tr>
<tr><td>2</td><td>Explain EM algorithm for Gaussian Mixture Models in detail. Derive E-step and M-step equations. Compare GMM with K-means clustering.</td><td>CO4</td><td>L3</td><td>15</td></tr>
<tr><td>3</td><td>Construct a dendrogram using single linkage and Complete Linkage for the following distance matrix:
<table>
<tr><td></td><td>A</td><td>B</td><td>C</td><td>D</td></tr>
<tr><td>A</td><td>0</td><td>2</td><td>6</td><td>10</td></tr>
<tr><td>B</td><td>2</td><td>0</td><td>5</td><td>9</td></tr>
<tr><td>C</td><td>6</td><td>5</td><td>0</td><td>4</td></tr>
<tr><td>D</td><td>10</td><td>9</td><td>4</td><td>0</td></tr>
</table></td><td>CO4</td><td>L4</td><td>15</td></tr>
<tr><td>4</td><td>An image processing company wants to compress grayscale images using PCA. Given 2D dataset:
<table>
<tr><td>Pixel 1</td><td>Pixel 2</td></tr>
<tr><td>2</td><td>1</td></tr>
<tr><td>3</td><td>2</td></tr>
<tr><td>4</td><td>3</td></tr>
<tr><td>5</td><td>4</td></tr>
</table>
a) Compute mean vector.
b) Compute covariance matrix.
c) Find eigenvalues and eigenvectors.
d) Reduce data to 1D using largest eigenvalue.</td><td>CO4</td><td>L4</td><td>15</td></tr>
<tr><td>5</td><td>A retail company wants to cluster customers based on Annual Income (in ₹ lakhs) and Spending Score. The dataset consists of</td><td>CO4</td><td>L4</td><td>15</td></tr>
</table>

Customer	Income	Score
C1	2	3
C2	3	4
C3	8	7
C4	9	8

Initial centroids:
Cluster 1 → (2,3)
Cluster 2 → (8,7)

a) Compute Euclidean distance matrix. (5 Marks)
b) Perform one iteration of K-means. (6 Marks)
c) Compute new centroids. (4 Marks)

UNIT-5: APPLICATION PROBLEMS

The case studies- churn analysis and prediction using Cox-proportional models, and churn prediction techniques. Credit card fraud analysis with a focus on handling imbalanced data and neural networks. Sentiment analysis- Sentiment mining from the New York Times are addressed using similarity measures like cosine similarity, chi-square, and N-grams. Part-of-speech tagging, stemming, chunking - sales funnel analysis, A/B testing, and campaign effectiveness. Web page layout effectiveness - recommendation systems with collaborative filtering - customer segmentation strategies and lifetime value- portfolio risk conformance and optimization, and Uber alternative routing with graph construction and route optimization.

PART-A (2-Marks)

S.NO	QUESTIONS	CO	RBT LEVEL
1	Compute conversion rates and Standard Error for A/B testing, Version A: 500 users → 50 conversions Version B: 500 users → 65 conversions	CO5	L2
2	Two documents using Term Frequency Vectors where D_1= (3,2,0,5), D_2 =(1,0,2,4) **a)** Compute Cosine Similarity between D_1 and D_2	CO5	L1
3	Mention two techniques to handle imbalanced datasets.	CO5	L2
4	State the assumption of the Cox Proportional Hazards model.	CO5	L1
5	Define N-grams.	CO5	L1
6	What is recommendation system sparsity problem?	CO5	L1
7	What is chi-square feature selection used for?	CO5	L1

8	Define hazard ratio in Cox model.	CO5	L1
9	Define A/B Testing?	CO5	L1
10	What is portfolio diversification?	CO5	L1
11	If shortest path cost is 10 and one edge increases by 3, what is new cost?	CO5	L2
12	Why is covariance important in portfolio optimization?	CO5	L1
13	If difference in proportion = 0.03 and standard error = 0.01, compute z-value.	CO5	L2
14	How many bigrams are generated from the sentence: "Data analytics improves decisions"?	CO5	L1
15	If $\beta = 0.003$ and the difference in monthly charges between two customers is 100, compute the hazard ratio	CO5	L2
16	A dataset has 5,000 transactions with 100 fraud cases. Compute fraud percentage.	CO5	L1
17	Compute the Cosine similarity for the vectors: (1,1) and (1,0).	CO5	L2
18	In A/B testing, what does a p-value less than 0.05 indicate?	CO5	L1
19	Why does cosine similarity ignore document length?	CO5	L1
20	If row total = 40, column total = 50, and grand total = 200, compute expected frequency.	CO5	L2

	PART-B (13- Marks)	CO	RBT Level	Marks
1	Consider a dataset, which is highly imbalanced (5% churn). Why is accuracy misleading? Compute precision, recall, F1 if: TP=30, FP=70, FN=20, TN=880. Which metric is more important for churn prediction?	CO5	L2	13
2	a)Two documents are represented as TF vectors: D1 = (2,1,0), D2 = (1,1,1) Compute cosine similarity and Interpret similarity score. Justify the normalization of vectors? b) Using chi square test, construct contingency table and compute chi-square statistic for the given data Word "excellent" appears: Positive docs = 40/100 Negative docs = 5/100 Justify why high chi-square implies strong sentiment signal?	CO5	L3	13
3	Discuss their role in sentiment analysis.	CO5	L2	13
4	Discuss evaluation metrics for churn prediction models.	CO5	L2	13

5	Explain the working of the Cox proportional hazards model for customer churn prediction**.**	**CO5**	**L2**	**13**
6	Compare and contrast user-based and item-based collaborative filtering.	**CO5**	**L2**	**13**
7	Compare Cosine Similarity and Chi-Square techniques in the context of sentiment analysis of large news datasets like the New York Times. Evaluate their effectiveness, computational complexity, and suitability for real-time sentiment mining.	**CO5**	**L2**	**13**
8	Why is alternative routing important in Uber systems? Explain how traffic congestion and roadblocks affect route selection.	**CO5**	**L2**	**13**
9	Discuss techniques to handle imbalanced datasets.	**CO5**	**L2**	**13**
10	Discuss the concept of N-grams in text mining. Explain how unigram, bigram, and trigram models help in extracting meaningful patterns from New York Times articles for sentiment classification.	**CO5**	**L2**	**13**
11	Explain collaborative filtering (user-based and item-based).	**CO5**	**L2**	**13**
12	Explain Part-of-Speech tagging, stemming, and chunking with example	**CO5**	**L2**	**13**
13	Explain how cosine similarity and chi-square statistics are used in sentiment mining of news articles.	**CO5**	**L3**	**13**
14	Design a statistical framework for evaluating two campaign versions.	**CO5**	**L2**	**13**
15	Explain how these improve sentiment mining and text classification performance. Discuss a) Tokenization b) Stemming c) POS tagging **d)** Chunking	**CO5**	**L2**	**13**
16	Explain how Customer Lifetime Value (CLV) is calculated and used for segmentation. Discuss a) CLV formula b) Retention rate impact c) RFM analysis d) Strategic marketing decisions	**CO5**	**L2**	**13**
17	Compare user-based and item-based collaborative filtering. Discuss: a) Cosine similarity computation b) Cold start problem	**CO5**	**L2**	**13**

	c) Sparsity issue d) Matrix factorization approach **e)** Scalability challenges			
18	Explain how businesses can use Customer Lifetime Value (CLV) to identify high-value customers and design targeted marketing strategies. Illustrate your answer with suitable examples and discuss the benefits and challenges involved in implementing segmentation and CLV analysis.	**CO5**	**L3**	**13**
19	Explain how a road network can be modeled using graph theory in ride-sharing applications like Uber also describe the process of graph construction , including nodes, edges, weighted edges and real – time data integration.	**CO5**	**L3**	**13**
20	Discuss challenges in fraud detection using neural networks under severe class imbalance.	**CO5**	**L2**	**13**
	PART-C (15-Marks)	**CO**	**RBT Level**	**Marks**
1	Consider the Credit Card Fraud Detection with Imbalanced Data for the given Dataset where Total transactions = 12,000, Fraudulent = 300, Genuine = 11,700 After oversampling fraud to 1,200: Neural network results: TP = 260, FP = 500, FN = 40, TN = 11,200 a) Compute class imbalance before and after oversampling. b) Compute Precision, Recall, F1-score. c) Compute Accuracy and comment why it may be misleading. d) Explain why Precision-Recall curve is preferred over ROC in imbalanced problems.	**CO5**	**L3**	**15**
2	An e-commerce company wants to improve conversions. Develop an integrated strategy combining sales funnel analysis, A/B testing, collaborative filtering recommendation systems, and customer lifetime value optimization.	**CO5**	**L3**	**15**
3	a) A bank wants to detect fraudulent credit card transactions with highly imbalanced data. Design a neural network-based fraud detection system. b) Explain preprocessing, sampling techniques, evaluation metrics, and challenges	**CO5**	**L4**	**15**
4	a) A telecom company experiences high churn. Design a complete churn prediction framework using Cox Proportional Hazard Model. **b)** Discuss assumptions, survival analysis concepts, and practical challenges	**CO5**	**L4**	**15**

5	Explain how Uber-like systems optimize routes based on multiple objectives such as travel time, fuel cost, distance, and driver availability. Discuss multi-objective optimization techniques and their role in improving efficiency and customer satisfaction. Illustrate your answer with a suitable diagram of graph representation.	**CO5**	**L3**	**15**

RBT LEVEL: L1- Remembering, L2 – Understanding, L3 – Applying, L4 – Analyzing, L5 – Evaluating, L6 – Creating